gt_dataset: hand-eye (marker only) vs diff-render (markerless), both scored against the ground truth (marker + 3D pose alignment)

First 5 frames

Frame	hand-eye trans. err (mm)	hand-eye rot. err (deg)	hand-eye IoU	diff-render trans. err (mm)	diff-render rot. err (deg)	diff-render IoU
0	19.13	0.00	0.056	4.87	6.91	0.783
1 *	19.13	0.00	0.000	4.66	6.91	0.753
2	19.13	0.00	0.002	4.26	6.91	0.737
3	19.13	0.00	0.008	4.45	6.91	0.678
4	19.13	0.00	0.000	4.62	6.91	0.673
Mean	**19.13**	**0.00**	**0.013**	**4.57**	**6.91**	**0.725**

Whole calibration, all 12 frames

Calibration	mean clevis displacement (mm)	rotation err (deg)	base-origin err (mm)
PnP init (no prior)	26.63	17.57	40.09
hand-eye (marker only)	19.13	0.00	19.13
diff-render (markerless)	4.43	6.91	10.92

trans. err = distance between the clevis origin (pitch pin) under the calibration and under the ground truth; rot. err = geodesic angle between the two clevis orientations (equal to the calibration's rotation error, so the same in every frame); IoU = overlap of the clevis silhouettes rendered at the calibration and at the ground truth. * = frame dropped by the diff-render outlier rejection (still scored). Diff-render inputs: rectified images, camera/stereo calibration, joint.stl, PSM kinematics -- no ground truth, marker calibration or initial pose.

```
================================================================================================================
    hand_eye_diff_render_comparison.py — markerless hand-eye calibration by differentiable rendering
================================================================================================================
Dataset   : /home/aseshad4/hand_eye_collection_scripts/gt_dataset  (12 frames; tip found in 12, 7 kept after outlier rejection)
Generated : 2026-10-26T18:43:47   fit runtime n/a (report rebuilt with --reuse-fit)
Ground truth = marker registration + 3D pose alignment (cam_T_base_post_alignment); used ONLY for scoring.
DR inputs   = rectified right+left images, intrinsics / rect_T_cam_raw / stereo P, joint.stl, PSM kinematics.
              No marker calibration, no ground truth, no initial pose, no annotation.
Pipeline    = stereo tool box -> GrabCut -> PnP init -> 4 DR passes (masks re-seeded from the
              DR estimate after pass 1) -> outlier rejection + refit.

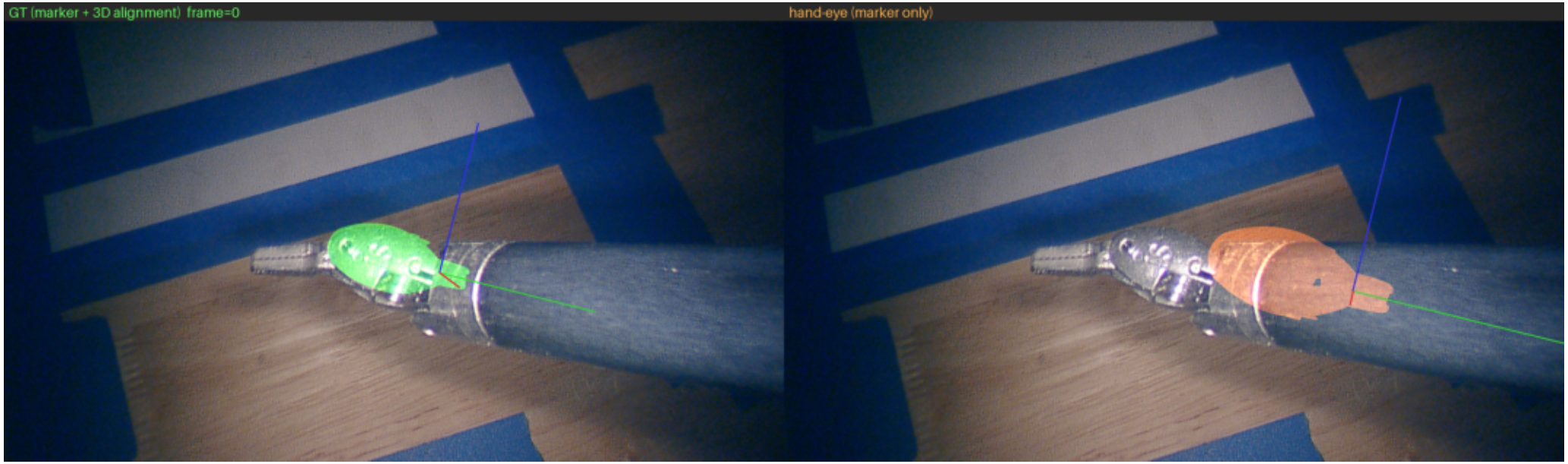
Whole calibration vs ground truth (all frames):
                            clevis displ. (mm)  rotation (deg)  base origin (mm)
  PnP init (no prior)                    26.63           17.57             40.09
  hand-eye (marker only)                 19.13            0.00             19.13
  diff-render (markerless)                4.43            6.91             10.92

First 5 frames (clevis pose vs its pose under the ground truth; IoU of the rendered silhouettes):
  frame kept    hand-eye (mark t(mm)  rot(deg)    IoU   diff-render (m t(mm)  rot(deg)    IoU
  ------------------------------------------------------------------------------------------
      0    y                   19.13      0.00  0.056                   4.87      6.91  0.783
      1    n                   19.13      0.00  0.000                   4.66      6.91  0.753
      2    y                   19.13      0.00  0.002                   4.26      6.91  0.737
      3    y                   19.13      0.00  0.008                   4.45      6.91  0.678
      4    y                   19.13      0.00  0.000                   4.62      6.91  0.673
  ------------------------------------------------------------------------------------------
   mean                        19.13      0.00  0.013                   4.57      6.91  0.725
  (t = clevis origin / pitch pin; rotation is the calibration's rotation error, the same for every frame)

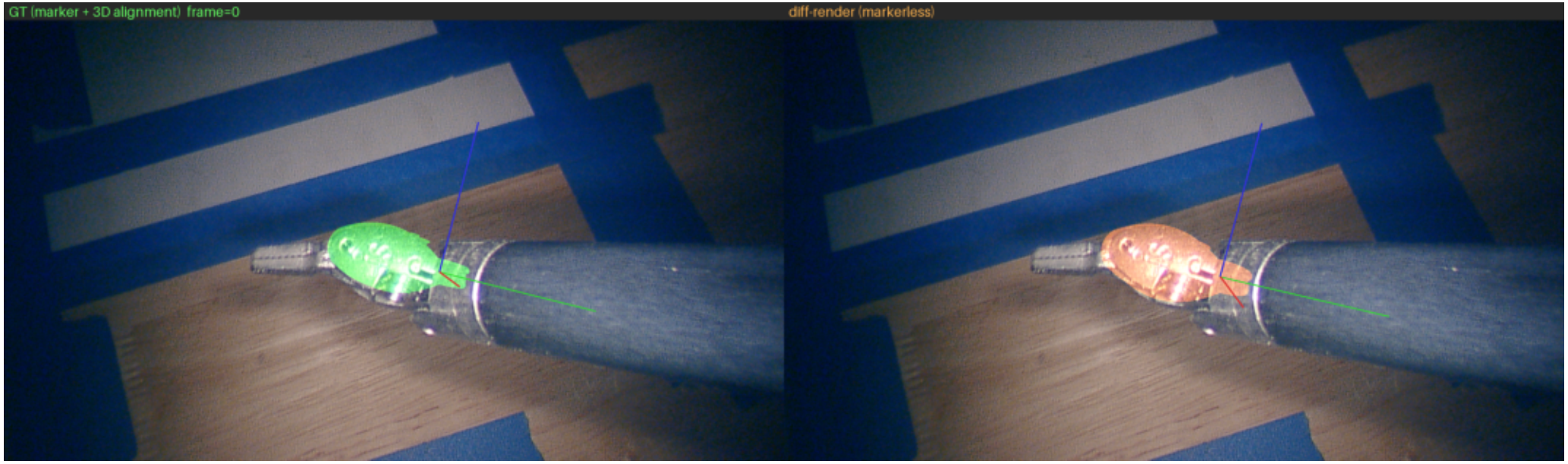
cam_raw_T_base (m), diff-render:
  [ 0.434501 -0.213425  0.87502   0.051076]
  [-0.246257  0.90635   0.343348 -0.003901]
  [-0.866352 -0.364665  0.341252 -0.015199]
  [0. 0. 0. 1.]
================================================================================================================
```

frame_0

hand-eye (marker only): translation error 19.13 mm rotation error 0.00 deg IoU 0.056

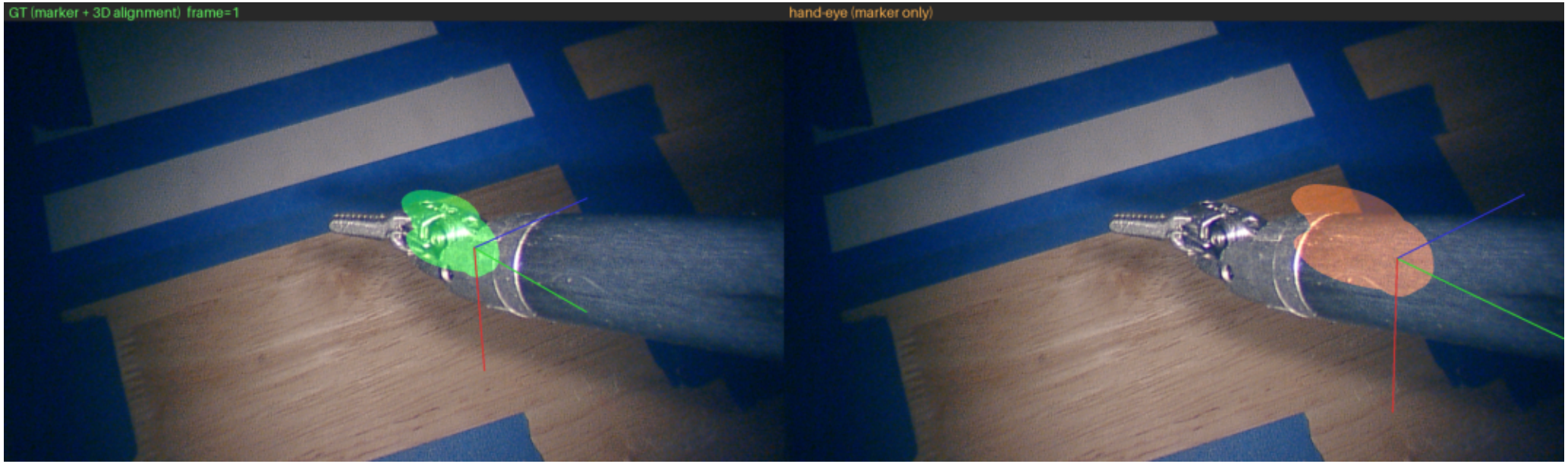


diff-render (markerless): translation error 4.87 mm rotation error 6.91 deg IoU 0.783

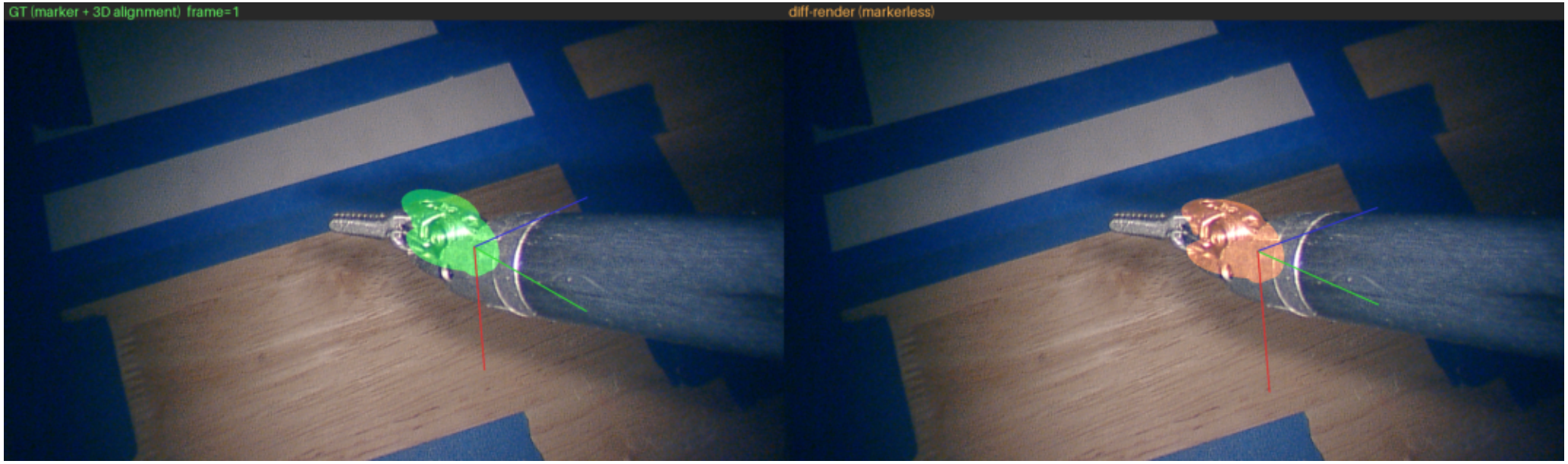


frame_1 (dropped by the diff-render outlier rejection; still scored)

hand-eye (marker only): translation error 19.13 mm rotation error 0.00 deg IoU 0.000

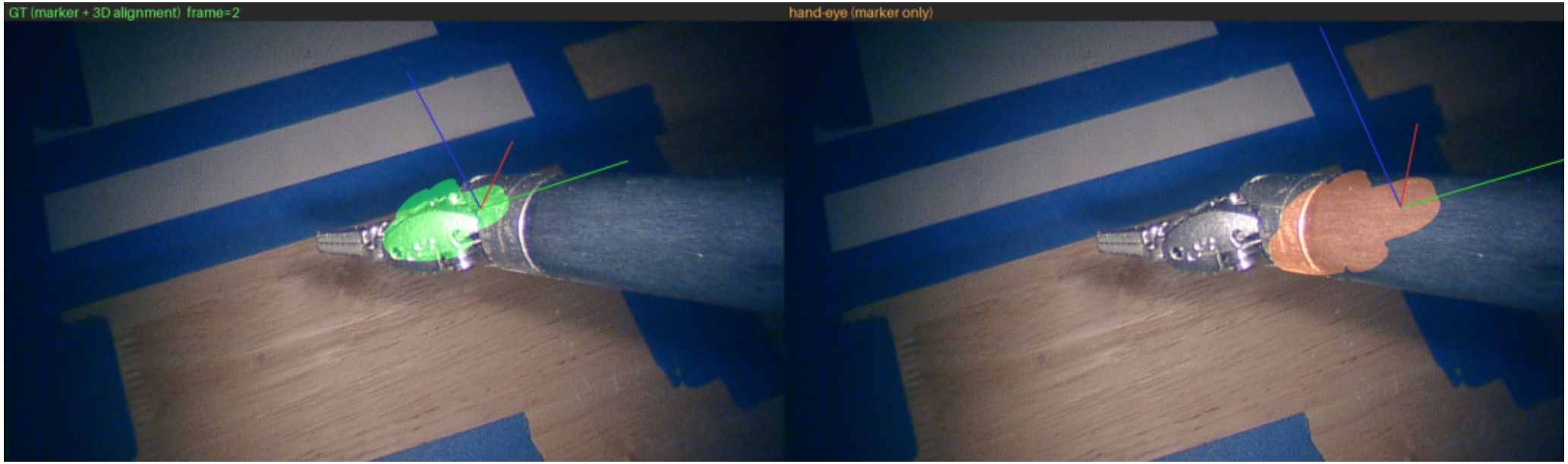


diff-render (markerless): translation error 4.66 mm rotation error 6.91 deg IoU 0.753

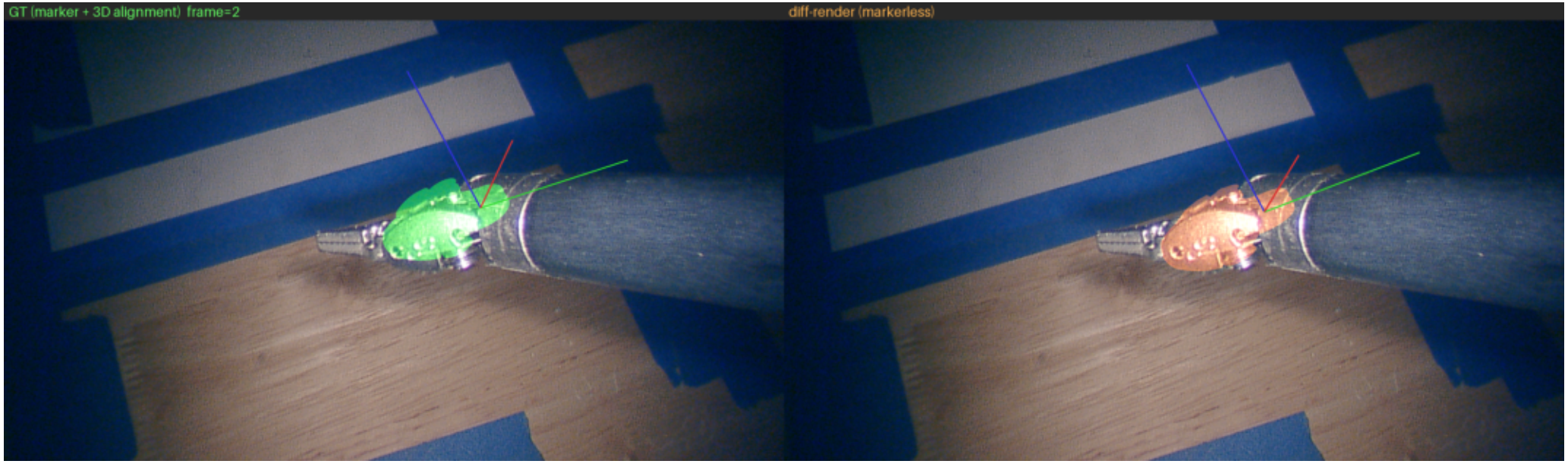


frame_2

hand-eye (marker only): translation error 19.13 mm rotation error 0.00 deg IoU 0.002

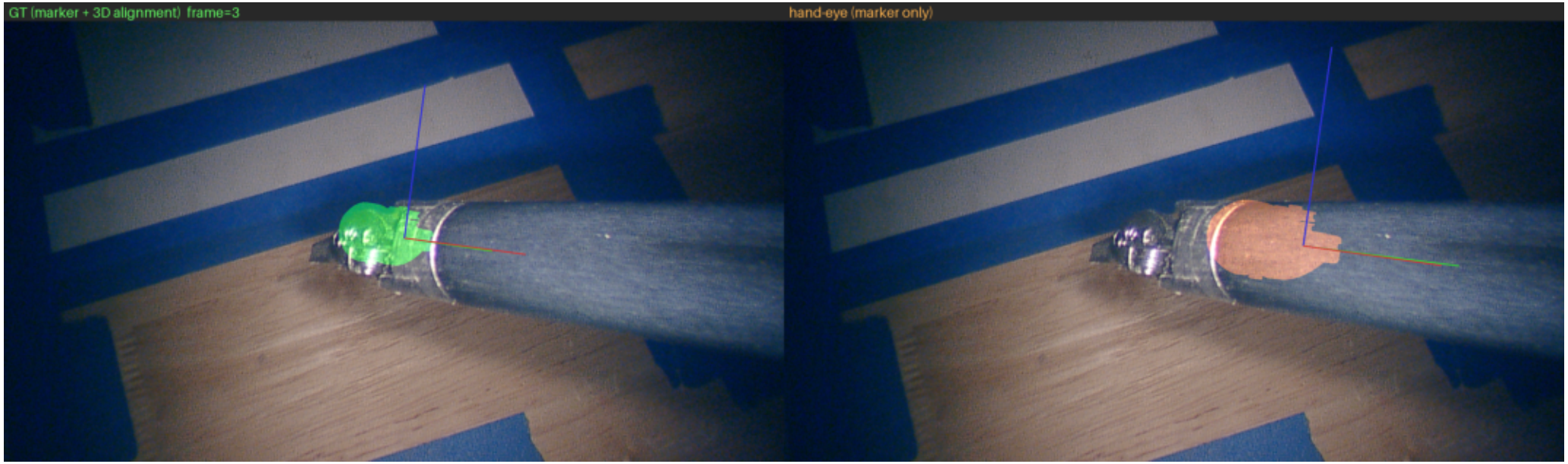


diff-render (markerless): translation error 4.26 mm rotation error 6.91 deg IoU 0.737

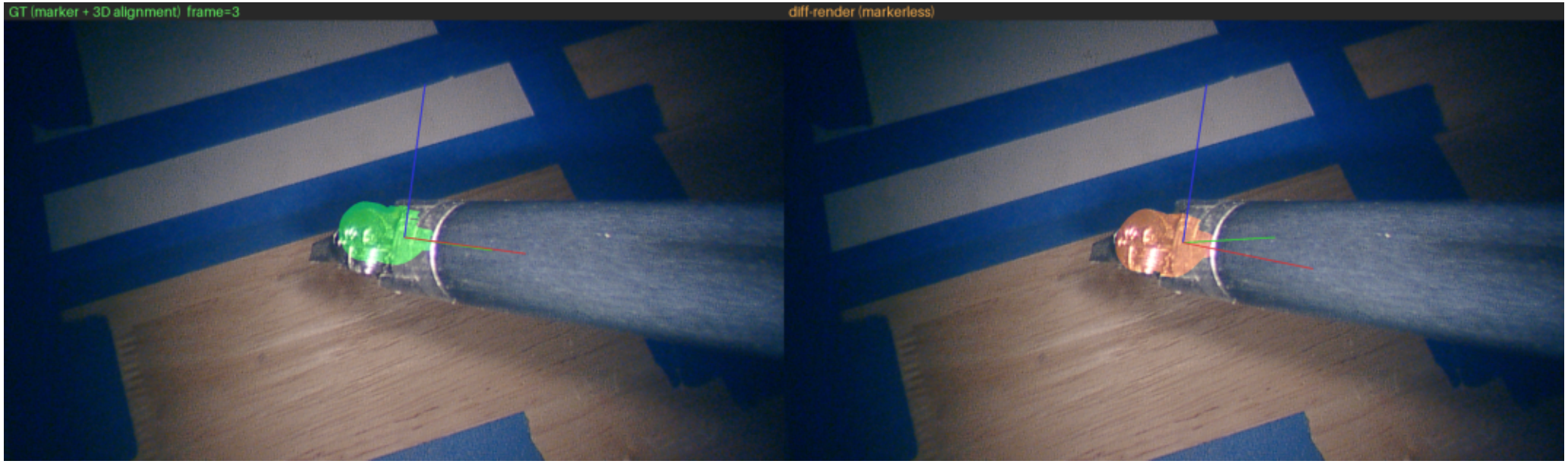


frame_3

hand-eye (marker only): translation error 19.13 mm rotation error 0.00 deg IoU 0.008

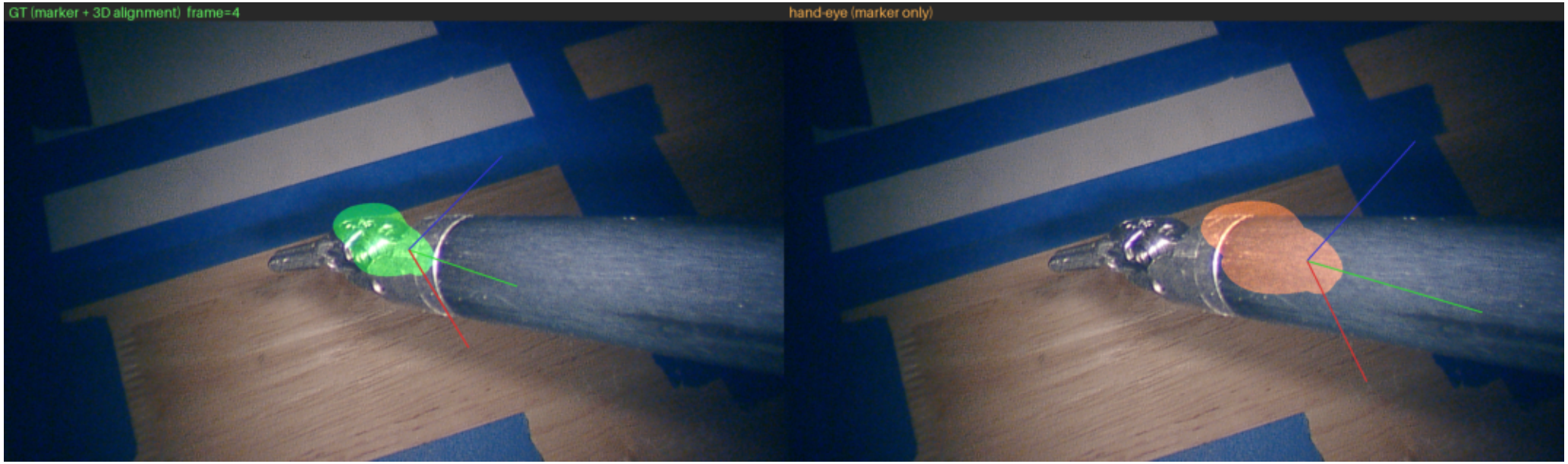


diff-render (markerless): translation error 4.45 mm rotation error 6.91 deg IoU 0.678



frame_4

hand-eye (marker only): translation error 19.13 mm rotation error 0.00 deg IoU 0.000



diff-render (markerless): translation error 4.62 mm rotation error 6.91 deg IoU 0.673

